Comprehensive Study Guide: Lecture 5 – Data Splitting

1. Core Principles of Generalization & Overfitting

- **Goal of Supervised Learning**: Build a predictive model that generalizes well to unseen future data, rather than simply memorizing the training dataset.
- **Overfitting**:
  - Occurs when a model learns the random noise or specific quirks of the training sample instead of the true underlying function $f(x)$.
  - **Symptom**: High accuracy / low error on training data, but poor accuracy / high error on test or validation data.
- **In-Sample vs. Out-of-Sample Error**:
  - **In-Sample Error (E_{train})**: Evaluation metric computed on the exact data used for parameter estimation.
  - **Out-of-Sample Error (E_{test})**: Evaluation metric computed on unseen data, representing true generalization performance.

2. The i.i.d. Assumption

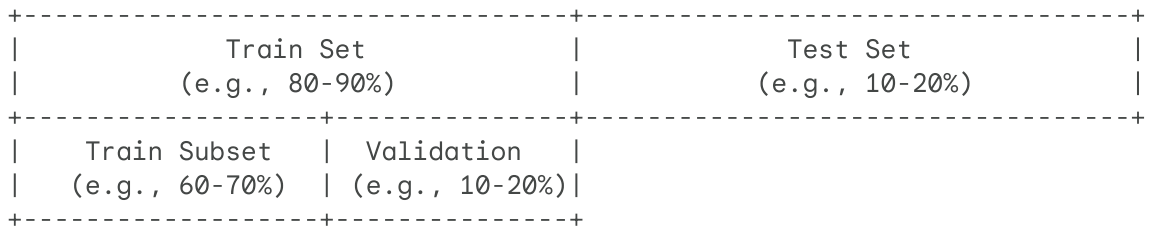
Most standard supervised learning algorithms rely on the assumption that dataset samples are **Independent and Identically Distributed (i.i.d.)**:

1. **Identically Distributed**:
   - All data samples (train, validation, and test) are drawn from the same underlying probability distribution $P(X, Y)$.
   - *Exam Note*: If the distribution shifts over time or across domains (data drift), model performance on test data degrades.
2. **Independent**:
   - The selection or realization of one sample provides zero statistical information about another sample.
   - *Violations*: Time series data (temporal autocorrelation), spatial data, grouped/clustered observations (e.g., multiple measurements from the same patient).

3. Dataset Splitting Workflow

To properly evaluate and tune models, the complete dataset D must be partitioned into distinct subsets:





Roles of the Subsets

Dataset Subset	Role & Usage	Permissible Operations
Training Set	Model parameter estimation (e.g., calculating weights, fitting decision trees).	Model `fit()`, parameter updating.
Validation Set	Hyperparameter tuning, model selection, and early stopping.	Evaluating model variations to pick optimal parameters.
Test Set	Final unbiased evaluation of the selected model's performance on unseen data.	Model `predict()` **ONLY** at the very end.

4. Preventing Data Leakage

Data Leakage occurs when information from outside the training dataset (such as validation or test data) is inadvertently used to train the model or engineer features.

Critical Rules

1. **Feature Scaling / Preprocessing**:
   - Always calculate statistics (e.g., mean μ, standard deviation σ, min, max) **ONLY** using the training fold.
   - Apply these fitted parameters to scale the validation and test sets:

$$x_{\text{scaled}} = \frac{x - \mu_{\text{train}}}{\sigma_{\text{train}}}$$

   - *Common Pitfall*: Calling `fit_transform()` on the entire dataset prior to splitting leads to overly optimistic test performance.
2. **Golden Rule**: Never inspect or use the test set during hyperparameter selection, feature engineering, or model architecture choice.

5. Cross-Validation Strategies

5.1 K-Fold Cross-Validation

- **Procedure**:
  1. Randomly divide the training dataset into K equal-sized folds.
  2. For fold $i = 1, \ldots, K$:
     - Train the model on $K - 1$ folds.
     - Validate the model on fold i to get score S_i.
  3. Average the validation performance: $\bar{S} = \frac{1}{K} \sum_{i=1}^{K} S_i$.
- **Advantages**:
  - Reduces variance in performance estimation compared to a single train/validation split.
  - Utilizes 100% of the training data for both training and validation across the iterations.

5.2 Leave-One-Out Cross-Validation (LOOCV)

- **Special Case**: K-Fold CV where $K = N$ (total number of samples).
- **Pros**: Unbiased estimator of test error.
- **Cons**: Computationally expensive (N models trained); high variance of the performance estimator because training sets share $N - 1$ samples.

5.3 Stratified K-Fold Cross-Validation

- **Problem with Standard K-Fold**: Random splitting can create folds with uneven target class distributions, especially in imbalanced datasets (e.g., 98% negative / 2% positive).
- **Solution**: Stratified K-Fold enforces that each fold maintains approximately the same class proportions as the entire dataset.
- **Usage**: Always recommended for classification tasks with imbalanced target variables.

6. Key `scikit-learn` Syntax Reference

```
from sklearn.model_selection import train_test_split, KFold, StratifiedKFold

# 1. Train / Test Split
X_train, X_test, y_train, y_test = train_test_split(
    X, y,
    test_size=0.2,
    random_state=42,
    stratify=y  # Essential for imbalanced classification
)

# 2. Stratified K-Fold
skf = StratifiedKFold(n_splits=5, shuffle=True, random_state=42)
for train_idx, val_idx in skf.split(X_train, y_train):
```

```
X_tr, X_val = X_train.iloc[train_idx], X_train.iloc[val_idx]
y_tr, y_val = y_train.iloc[train_idx], y_train.iloc[val_idx]
```

Quick Exam Self-Check List

- [ ] Can you define **identically distributed** vs. **independent** in your own words?
- [ ] Do you know why `random_state` is set during splitting? *(Answer: To ensure reproducibility across runs.)*
- [ ] Why should the test set never be touched during feature engineering or hyperparameter tuning?
- [ ] When should you use **Stratified K-Fold** over standard K-Fold?
- [ ] What is the main drawback of **Leave-One-Out Cross-Validation (LOOCV)**?

```
# Next Steps for Review
**Focus on your missed quiz concepts**: Review the **i.i.d. assumption** (speci
Feel free to ask if you would like me to clarify any specific point or work thr
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The primary objective of data splitting is to reliably estimate how a machine learning model will perform on previously unseen data. Evaluating a model on the same data used to train it leads to overfitting and overly optimistic performance estimates.

Key Conceptual Metaphor — Data Splitting as an Academic Course:

Training set: The homework assignments and practice problems you study with to learn the material.

Validation set: Practice quizzes used to test and evaluate different study techniques (hyperparameter tuning and model selection).

Test set: The official final exam. If you look at the exam questions while studying or tuning your answers, your high grade won't reflect your actual understanding of the material.